Super_Store_Orders

E2
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ASK: Define the problem, business task, and list objectives, stakeholders and questions

- <u>Problem:</u> The super store is facing challenges optimizing profit margins across product lines and international markets. Certain regions, products and shipping methods are underperforming and creating losses.
- <u>Business Task:</u> Analyse historical sales data from 2011 to 2014 to identify key profitability and sales growth drivers.
- <u>Objectives:</u>

- Identify seasonal sales patterns and year-over-year growth to forecast future demand.
- Categorize products into "Winners" (high profit) and "Underperformers" (low profit/loss) to refine the inventory mix.
- Map global sales to find high-potential markets and regions requiring operational improvements.
- Evaluate the impact of shipping modes and order priorities on net profit (the bottom line).
- Determine the "sweet spot" for discounts that drives volume without eroding total profit.

- Stakeholders: Executive Leadership (CEO/CFO), Sales Managers, Product/Category Managers, Logistics & Supply Chain Team, and Marketing Team
- Questions: What is the total sales and profit trend over the four-year period?
    - Which product categories and regions are the most and least profitable?
    - Is there a correlation between high discounts and low profit?
    - Which shipping modes are most commonly used and how do they impact shipping costs?

Prepare: List data source, structure, ROCCC evaluation, transformations, constraints and integrity

- Data source: This dataset was made available by Aditi Saxena and can be accessed through the following link: https://www.kaggle.com/datasets/aditisaxena20/superstore-sales-dataset
- ROCCC Checklist: Does this dataset ROCCC or not?
    - Reliable: This dataset scores medium on reliability, it is well known and does not contain any null values, but it is likely synthetic and does not represent a specific real-world company's current books.
    - Original: This dataset scores low on dataset originality. It was made available by a secondary source, it was originally created for Tableau/Microsoft training and has been re-uploaded to Kaggle by multiple users.
    - Comprehensive: This dataset is highly comprehensive; it contains every dimension needed for a retail analysis.
    - Current: The data is not current and is considered stale since it is from 2011 to 2014 which is more than 10 years back.
    - Cited: This dataset scores high on citations, it is one of the most famous datasets in data science
- Transformations: This dataset was downloaded as a .zip file which contains the .csv file that will be imported in the process phase.
- Data structure: The dataset has 51,290 rows and 21 columns
    - columns: order_id, order_date, ship_date, ship_mode, customer_name, segment, state, country, market, region, product_id, category, sub_category, product_name, sales, quantity, discount, profit, shipping_cost, order_priority, year.
    - Data types: String, String, String, String, String, String, String, String, String, String, String, String, String, String, String, Integer, Float, Float, Float, String, Integer.

- Constraints: The order_id is not unique, one order can contain multiple products. A primary key needs to be generated either using the combination of order_id and product_id or a generated Identity column to act as the unique identifier. The sales column is imported as a string, it must be converted to a FLOAT or DECIMAL for calculation. Dates in this CSV use a mix of formats (e.g., 1/1/2011 vs 18-06-2014). SQL Server requires a standardized YYYY-MM-DD format to perform time-series analysis correctly. Profit contains negative values (losses). This is a constraint for certain types of visualizations (like Pie Charts) but essential for "Identify the problem" analysis. Some "States" may exist in multiple countries, use the Country and State columns together in Power BI maps to ensure dots appear in the correct location.
- Data Integrity: This data does not include Personal Identifiable Information like Phone numbers, Identity numbers or credit card numbers, so users' privacy is protected. The customer names are likely synthetic as well.
- Tool selection: SQL Server and Power BI, I chose SQL Server to efficiently cleans and processes large datasets and Power BI to transform that data into interactive dashboards.

Process: Load, Clean and Transform the dataset

To begin the data processing stage of the project, I first created a dedicated database in **Microsoft SQL Server Management Studio** to store and manage the dataset. After creating the database, I used the **Flat File Import Wizard** to upload the raw .csv dataset into a new table called **SuperStoreOrders**. During the import process, I initially allowed SQL Server to infer the column structure and generate the table automatically. Once the data was loaded, I reviewed the schema to verify column names, data types, and column lengths.

While inspecting the dataset, I discovered that several columns required additional processing before the data could be used for analysis. Financial columns such as **sales, profit, shipping_cost, and discount** were initially stored as text values due to formatting inconsistencies in the source file. Similarly, the **order_date** and **ship_date** columns required conversion from string format to proper SQL date data types. To resolve these issues, I cleaned the data by removing formatting characters, validating numeric values, and converting the columns into appropriate data types. I also verified that the transformations were successful by performing validation checks on date ranges, numeric ranges, and record counts. This ensured that the dataset was accurate, consistent, and ready for analytical queries.

Queries Executed in the Process Phase

The following SQL queries were executed as part of the data preparation process:

- Create database query
- Schema verification query to inspect column names and data types
- Data cleaning queries to remove formatting characters (such as commas) from numeric columns
- Validation queries using TRY_CONVERT to identify non-numeric values
- Column conversion queries to change financial columns to DECIMAL data types

- Date conversion queries to convert order_date and ship_date to DATE format
- Data validation queries to confirm correct numeric ranges and date ranges
- Final schema verification query to confirm that all columns were successfully processed

Click this link to access queries:
https://github.com/e2skm/Data-Analytics-with-SQL/tree/main/super_store_orders/process_queries

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Analyse: Explore, aggregate and model the data

After completing the data processing stage, I moved into the analysis phase where I used SQL queries to explore the dataset and identify meaningful patterns in sales performance. The objective of this phase was to transform the cleaned dataset into insights that could support business decision-making. Using structured queries in **Microsoft SQL Server Management Studio**, I calculated key performance metrics, evaluated profitability, and identified sales trends across different regions, product categories, and time periods.

I began by generating a **high-level financial snapshot** to understand the overall performance of the business. This included calculating total sales, total profit, average discount, and profit margin percentage. Next, I analyzed **sales trends over time** to identify seasonal patterns in revenue. I then grouped and aggregated the data by different business dimensions such as **country, region, category, and product**, which helped reveal which markets and products contributed most to overall revenue and profit. Finally, I conducted a **hypothesis test on discount levels** to evaluate how different discount tiers affected profitability. These analytical queries helped uncover key insights that were later visualized in the Power BI dashboards created during the Share phase.

<u>Queries Executed in the Analyse Phase</u>

The following SQL queries were executed to perform the analysis:

- High-level financial snapshot query calculating total sales, total profit, and profit margin
- Revenue aggregation queries grouped by **year and month** to identify sales trends
- Revenue and profit aggregation queries grouped by **country and region**
- Product and category performance queries identifying top-performing products
- Discount analysis queries grouping orders by discount tiers
- Hypothesis testing queries evaluating the relationship between discount levels and profitability

Click this link to access queries:
https://github.com/e2skm/Data-Analytics-with-SQL/tree/main/super_store_orders/analyse_queries

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Share Phase: Create Dashboards to communicate data insights

In the Share phase, I translated the analytical results into clear and accessible visual insights using **Microsoft Power BI**. After exporting the processed dataset from **Microsoft SQL Server Management Studio** into a CSV format, I imported the data into Power BI to design interactive dashboards that communicate the key findings of the analysis. The goal of this phase was to transform the SQL-based insights into visualizations that non-technical stakeholders could easily understand and use for decision-making.

During the dashboard development process, I first verified the data model and ensured that all columns had the correct data types. I then created calculated columns and measures such as **Profit Margin %**, **Average Discount %**, and **Discount Groups** to support meaningful visual analysis. Using these measures, I designed a set of dashboards that highlight important performance indicators, trends, and relationships within the dataset. Throughout the design process, I focused on simplicity, clarity, and accessibility by using appropriate chart types, clear labeling, and a consistent color theme to ensure that the key insights stood out.

In total, I created **three dashboards**, each designed for a different stakeholder perspective. The **Executive Overview dashboard** presents high-level business performance indicators such as total sales, total profit, profit margin, and monthly revenue trends. The **Sales Performance dashboard** focuses on identifying revenue drivers by highlighting top-performing products, sales by region, and sales by customer segments. The **Profitability and Discount Analysis dashboard** explores the relationship between discount strategies and profit margins, helping identify areas where pricing strategies may impact profitability.

These dashboards visually communicate the main findings of the analysis, including seasonal revenue trends, product and regional performance differences, and the impact of discounts on profitability. By presenting the insights through interactive visuals and clear metrics, the dashboards enable stakeholders to quickly understand the business performance and identify opportunities for improvement.

<u>Visualizations Created in the Share Phase</u>

The following visualizations were created in Power BI to communicate the analysis results:

- **KPI Card visuals** displaying Total Sales, Total Profit, Profit Margin %, and Average Discount %
- **Monthly Revenue Line Chart** showing sales trends over time
- **Top Products Bar Chart** highlighting the highest revenue-generating products
- **Regional Sales Performance Chart** comparing sales across regions
- **Sales by Segment Chart** illustrating revenue distribution across customer segments
- **Discount Group Analysis Chart** evaluating the impact of discounts on profitability

These visualizations allowed the analysis results to be communicated clearly and effectively, supporting the final stage of the project where data-driven recommendations were developed.

Click this link to access queries:
https://app.powerbi.com/links/CcUg8fv1oG?ctid=3df74539-9453-4d03-bb9d-b9102cb9ce9c&pbi_source=linkShare&bookmarkGuid=ec478b57-c4d2-464e-88d5-105cb11d556a

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ACT: Provide Recommendations

1. Prepare Inventory and Marketing for High-Demand Months: The Monthly Revenue Trend shows clear seasonal peaks, with November consistently generating the highest sales. This indicates strong demand during this period.
2. Review Discount Strategy to Protect Profit Margins: The Profitability and Discount Analysis dashboard shows that higher discount tiers are associated with lower profit margins. While discounts may increase sales volume, excessive discounting reduces overall profitability.
3. Focus Sales Efforts on High-Performing Products and Regions: The Sales Performance Dashboard shows that a small group of products and regions generate a large share of total revenue.